

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

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Section: B

1. TITLE

HTML5 Graphics and CSS3 Web Styling

2. AIM

To understand how CSS3 enhances the appearance of web pages and to explore HTML5 graphics using the Canvas element by designing a visually styled webpage.

3. OBJECTIVE

The objectives of this experiment are:

- To draw basic shapes and text on a webpage using the HTML5 Canvas element.
- To understand CSS3 syntax and different types of selectors.
- To apply colors, fonts, borders, margins and padding to page elements.
- To use CSS3 backgrounds, gradients, box-shadow, text-shadow and rounded corners.
- To implement hover effects on buttons and navigation links.

4. THEORY

The HTML5 <canvas> element provides a drawable region on a webpage that can be controlled using JavaScript. Its 2D context object exposes methods such as strokeRect(), arc(), moveTo()/lineTo() and fillText() to draw rectangles, circles, lines and text respectively.

CSS3 (Cascading Style Sheets, Level 3) is used to control the presentation of HTML elements. Selectors (element, class, id) target specific elements, while properties such as background, border-radius, box-shadow, text-shadow and linear-gradient() control their appearance. The :hover pseudo-class applies styles only while the mouse pointer is over an element, and combined with the transition property it produces smooth animated effects such as a card lifting or a button changing color.

5. ALGORITHM / PROCEDURE

- 1 Create a new HTML file and declare the HTML5 document type.
- 2 Add a <canvas> element inside the body and give it an id.
- 3 In a script block, get the 2D drawing context of the canvas using getContext("2d").
- 4 Draw a rectangle using strokeRect().
- 5 Draw a circle using arc() with beginPath() and stroke().
- 6 Draw a straight line using moveTo() and.lineTo().

- 7 Display the student's name using fillText().
- 8 Build the page layout with a header, navigation bar, three content cards and a footer.
- 9 Apply CSS3 background colors and a gradient background to the header.
- 10 Style the cards with borders, border-radius, padding, margin and box-shadow.
- 11 Add a text-shadow to the header title and apply different font families.
- 12 Add hover effects on the navigation links and on the buttons using :hover and transition.
- 13 Save the file, open it in a browser and verify the drawing and styling.

6. PROGRAM

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">

  <!-- Makes the webpage responsive on mobile devices -->
  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>My Styled Webpage</title>

  <!-- Connecting the external CSS file -->
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <!-- Header section -->
  <header>

    <h1>My Styled Webpage</h1>

    <p>
      Learning HTML5 Canvas and CSS3 Web Styling
    </p>

  </header>

  <!-- Navigation bar -->
  <nav>

    <a href="#home">Home</a>

    <a href="#canvas">Canvas</a>

    <a href="#cards">Services</a>

    <a href="#contact">Contact</a>

  </nav>

  <!-- Main content -->
  <main>

    <!-- Canvas section -->
    <section id="canvas">

      <h2>HTML5 Canvas Graphics</h2>

      <!-- Canvas drawing area -->
      <canvas
        id="myCanvas"
        width="700"
```

```

        height="350"
    >
        Your browser does not support HTML5 Canvas.
    </canvas>

</section>

<!-- Content cards section -->
<section id="cards">

    <h2>Our Learning Areas</h2>

    <div class="card-container">

        <!-- Card 1 -->
        <article class="card">

            <h3>HTML5</h3>

            <p>
                HTML5 is used to create the structure
                and content of a webpage.
            </p>

            <button>Learn HTML</button>

        </article>

        <!-- Card 2 -->
        <article class="card">

            <h3>CSS3</h3>

            <p>
                CSS3 is used to add colours, fonts,
                spacing, borders, and visual effects.
            </p>

            <button>Learn CSS</button>

        </article>

        <!-- Card 3 -->
        <article class="card">

            <h3>HTML5 Canvas</h3>

            <p>
                Canvas is used to draw shapes, text,
                graphics, animations, and games.
            </p>

            <button>Explore Canvas</button>

        </article>

    </div>

</section>

</main>

<!-- Footer section -->
<footer id="contact">

    <p>
        &copy; 2026 My Styled Webpage |

```

```

        Designed using HTML5 and CSS3
    </p>

</footer>

<!-- JavaScript for Canvas drawing -->
<script>

    // Get the canvas element
    const canvas =
        document.getElementById("myCanvas");

    // Create a 2D drawing context
    const context =
        canvas.getContext("2d");

    // -----
    // Draw a rectangle
    // -----

    context.fillStyle = "skyblue";

    context.fillRect(
        40,
        50,
        180,
        100
    );

    // -----
    // Draw a circle
    // -----

    context.beginPath();

    context.arc(
        350,
        100,
        50,
        0,
        2 * Math.PI
    );

    context.fillStyle = "orange";

    context.fill();

    // -----
    // Draw a line
    // -----

    context.beginPath();

    context.moveTo(
        450,
        200
    );

    context.lineTo(
        650,
        80
    );

    context.strokeStyle = "green";

    context.lineWidth = 5;
    context.stroke();

```

```

// -----
// Display name as text
// -----

context.font =
    "bold 30px Arial";

context.fillStyle =
    "purple";

context.fillText(
    "Kongarani Sirisha",
    220,
    300
);

</script>
</body>
<style>
/* -----
   Universal Selector
----- */

* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

/* -----
   Body Styling
----- */

body {

    font-family:
        Arial,
        Helvetica,
        sans-serif;

    background-color:
        #e1e5ec;

    color:
        #bf3838;

    line-height:
        1.9;
}

/* -----
   Header Styling
----- */

header {

    /* Gradient background */
    background:
        linear-gradient(
            90deg,
            #1e3c72,
            #71c1dc
        );

    color:
        rgb(237, 237, 237);

    text-align:
        center;

```

```
padding:
    35px;

/* Text shadow */
text-shadow:
    2px 2px 5px black;

}
```

```
header h1 {

    font-family:
        Georgia,
        serif;

    font-size:
        42px;

}
```

```
header p {

    font-size:
        20px;

    margin-top:
        10px;

}
```

```
/* -----
   Navigation Bar
   ----- */
```

```
nav {

    background-color:
        #111;

    text-align:
        center;

    padding:
        15px;

}
```

```
nav a {

    color:
        white;

    text-decoration:
        none;

    font-size:
        18px;

    margin:
        0 20px;

    padding:
        8px 12px;

    border-radius:
        6px;

    transition:
```

```

        0.3s;
    }

/* Navigation hover effect */
nav a:hover {
    background-color:
        rgb(255, 255, 255);

    color:
        #1e3c72;
}

/* -----
Main Section
----- */

main {
    width:
        90%;

    max-width:
        1100px;

    margin:
        30px auto;
}

section {
    margin-bottom:
        50px;
}

section h2 {
    text-align:
        center;

    font-family:
        Georgia,
        serif;

    font-size:
        30px;

    color:
        #1e3c72;

    margin-bottom:
        25px;
}

/* -----
Canvas Styling
----- */

canvas {
    display:

```

```

        block;

max-width:
    100%;

margin:
    auto;

background-color:
    white;

border:
    4px solid #1e3c72;

border-radius:
    15px;

padding:
    10px;

/* Box shadow */
box-shadow:
    0 6px 18px
    rgba(0, 0, 0, 0.25);
}

/* -----
Card Container
----- */

.card-container {

    display:
        flex;

    justify-content:
        center;

    gap:
        25px;

    flex-wrap:
        wrap;
}

/* -----
Content Cards
----- */

.card {

    width:
        300px;

    min-height:
        260px;

    background:
        linear-gradient(
            135deg,
            white,
            #dbeafe
        );

    border:
        2px solid #2a5298;

    border-radius:

```



```

        20px;

margin:
    10px;

padding:
    25px;

text-align:
    center;

/* Box shadow */
box-shadow:
    0 8px 18px
    rgba(0, 0, 0, 0.2);

transition:
    0.3s;
}

/* Card hover effect */
.card:hover {

    transform:
        translateY(-10px);

    box-shadow:
        0 14px 25px
        rgba(0, 0, 0, 0.3);
}

.card h3 {

    color:
        #1e3c72;

    font-size:
        25px;

    margin-bottom:
        15px;
}

.card p {

    font-family:
        Verdana,
        sans-serif;

    font-size:
        16px;

    margin-bottom:
        20px;
}

/* -----
   Button Styling
   ----- */

button {

    background-color:

```

```

        #1e3c72;

color:
    white;

font-size:
    16px;

padding:
    10px 18px;

border:
    none;

border-radius:
    25px;

cursor:
    pointer;

transition:
    0.3s;
}

/* Button hover effect */
button:hover {

    background-color:
        #f97316;

    transform:
        scale(1.1);

    box-shadow:
        0 5px 12px
        rgba(0, 0, 0, 0.3);
}

/* -----
   Footer Styling
   ----- */

footer {

    background-color:
        #111;

    color:
        white;

    text-align:
        center;

    padding:
        25px;

    font-size:
        17px;
}

</style>
</html>

```

7. CODE EXPLANATION

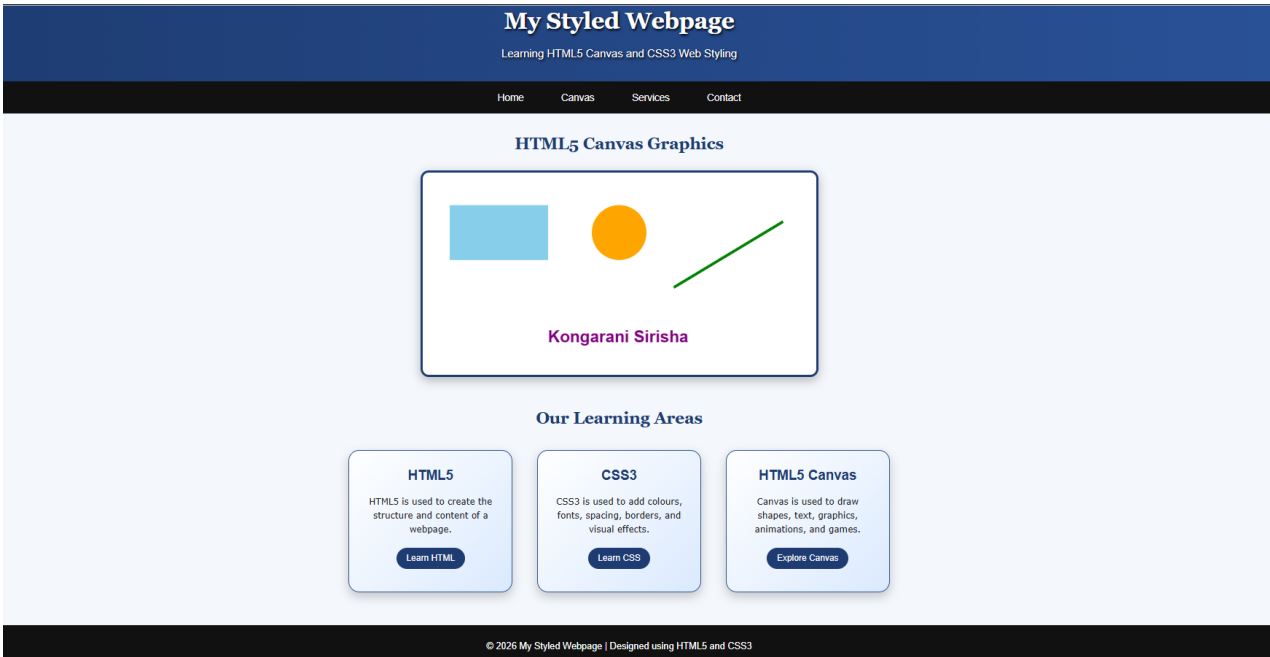
Code Explanation

Code	Explanation
<canvas id="myCanvas">	Reserves a drawable area on the page.
getContext("2d")	Returns the 2D drawing context used to draw shapes.
fillRect(x,y,w,h) / strokeRect(x,y,w,h)	Draws a filled or outlined rectangle at the given position and size.
arc(x,y,r,0,2*PI) + fill()/stroke()	Draws a circle of radius r centred at (x, y).
moveTo() / lineTo()	Draws a straight line between two points.
fillText()	Renders text (the student's name) onto the canvas.
linear-gradient()	Creates a smooth gradient background for the header and cards.
box-shadow	Adds a soft drop shadow around the canvas and content cards.
text-shadow	Adds a shadow behind the header title text.
border-radius	Rounds the corners of the canvas, cards and buttons.
:hover + transition	Animates nav links, cards and buttons smoothly when the mouse hovers over them.

8. TEST CASES AND EXECUTION DATA

Test Case	Action / Input	Expected Result	Actual Result	Status
TC-01	Load the page in browser	Header, navigation bar, canvas and cards display correctly	Displayed correctly	Pass
TC-02	Inspect the canvas area	Rectangle, circle, line and name text are visible	All four elements rendered	Pass
TC-03	Hover over a navigation link	Link background changes with a smooth transition	Background changed smoothly	Pass
TC-04	Hover over a content card	Card lifts up with an enhanced shadow	Lift and shadow effect worked	Pass
TC-05	Hover over a button	Button color changes and scales slightly	Color changed and button scaled	Pass
TC-06	Resize the browser window	Cards wrap responsively without overlapping	Cards wrapped correctly	Pass

9. OUTPUT



10. RESULT

Thus, a webpage titled "My Styled Webpage" was successfully developed, demonstrating HTML5 Canvas drawing (rectangle, circle, line and text) along with CSS3 styling features such as gradients, shadows, rounded corners and hover transition effects. The output was verified against the expected results for all test cases.